

**Department of Physics
University of Florida**

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PHY 2049

Final Exam

December 10th, 2024

On my honor, I have neither given nor received unauthorized aid on this examination

Name

Last: _____ First: _____ Signature: _____

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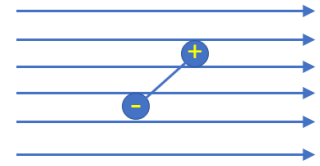
Directions

1. Print your name on this question sheet and sign it.
2. Code your 5-digit exam code (at the top of this question sheet) on your Scantron in columns 76–80.
3. Code your name and 8-digit UFID on the Scantron. **Darken all circles completely on the Scantron** using a #2 pencil or **dark blue** or **black** ink. Do not make any stray marks or the Scantron may not read properly.
4. Do all scratch work anywhere on this question sheet that you like. At the end of the exam, the question sheet and Scantron are both to be turned in. *No credit will be given without both Scantron sheet and question sheet.*
5. The answers are rounded, so choose the closest to exact. There is no penalty for guessing.

Correct answer marked with ♣

1. A small electric dipole is placed in a uniform electric field of direction as shown. The dipole is free to move. Its initial movement will be:

- (1) ♣ Clockwise rotation but no movement of its center of mass
 (2) Clockwise rotation but also movement of its center of mass to the right
 (3) Movement of its center of mass to the left but no rotation
 (4) Clockwise rotation but also movement of its center of mass to the left
 (5) No movement



The top of the dipole (positive charge) will be pulled to the right, and the bottom (negative) will be pulled left. That will twist the dipole clockwise. However, the net force on it will be zero so there will not be any movement of the center of mass (note that this is because it is in a uniform electric field, not one caused by a point charge).

2. Two charges, $+Q$ and $-2Q$, are placed on the x -axis at $x = 0$ and $x = 5$ respectively. Other than trivial answers of $\pm\infty$, where on the x axis is a location where the electric field is zero? Answer to the nearest integer.

- (1) ♣ -12 (2) -5 (3) 2 (4) 17 (5) -17

It's best to draw a graph. It should be clear that any small positive charge between the two fixed charges will be pushed by the $+Q$ to the positive x direction and pulled by the $-2Q$ in the same direction, so the electric field cannot be zero. On the other hand, at negative values of x , closer to the $+Q$ charge than the $-2Q$ charge, it is possible to have zero electric field. If the location is $-x$, then the following expression is the electric field in the $+x$ direction: $-kQ/x^2 + 2kQ/(x+5)^2 = 0$, $2/(x+5)^2 = 1/x^2$, $2x^2 = x^2 + 10x + 25$, $x^2 - 10x - 25 = 0$, ~ 12 , So, the electric field is zero at $x = -12$.

3. A long cylindrical shell of outer radius 4.0 cm and inner radius 2.0 cm has uniform volume charge density $\rho = 10 \mu\text{C}/\text{m}^3$. What is the magnitude of the electric field at radial distance $r = 3.0$ cm from the central axis of the shell, in kN/C ?

- (1) ♣ 9.4 (2) 7.3 (3) 5.2 (4) 3.1 (5) 0

This is Gauss' law for a cylindrically symmetric system. Here $\int \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enclosed}}}{\epsilon_0} = E2\pi r\ell$, where $r = 3.0$ cm and ℓ is the length of the cylinder. So we have to find Q_{enclosed} , which is $\rho(\pi r^2\ell - \pi R_{\text{in}}^2\ell)$, where $R_{\text{in}} = 2.0$ cm is the inner radius of the cylindrical shell. So $E = \frac{\rho(r^2 - R_{\text{in}}^2)/r}{2\epsilon_0} = 1.0 \times 10^{-5} [0.03 - (0.02)^2/0.03] / [2(8.85 \times 10^{-12})] = 9.4 \times 10^3 \text{ N/C}$. Without even remembering or understanding Gauss' Law, you can use the standard formula for a line charge as $E = \frac{\lambda}{2\pi\epsilon_0 r}$. Here λ is the charge per unit length inside $r = 3 \text{ cm}$ (and it is symmetric). That is $\lambda = \rho(\pi r^2 - \pi R_{\text{in}}^2)$.

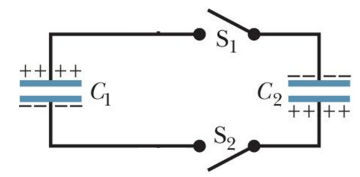
4. Two parallel, infinite, non-conducting planes, 1 and 2, are separated by 1.0 cm. Plane 1 is uniformly charged with surface charge density $\sigma_1 = +1.0 \mu\text{C}/\text{m}^2$; plane 2 is also uniformly charged but its surface charge density, σ_2 , is not known. What is σ_2 , in $\mu\text{C}/\text{m}^2$, if the electric-potential difference $V_2 - V_1$ between planes 2 and 1 is $+2.0 \text{ kV}$?

- (1) ♣ +2.5 (2) -1.9 (3) +3.2 (4) -4.3 (5) 0

The electric field from each plane is constant and given by $E = \frac{\sigma}{2\epsilon_0}$. Since plane 1 is positively charged, the electric field due to the charge on this plane points toward plane 2 in the space between the two plates. The surface charge density for plane 2 is not given, but since plane 2 has a higher electric potential than plane 1, its surface charge density should be larger than σ_1 , producing a larger field pointing toward plane 1 than the field, toward plane 2, due the charge on plane 1. This leaves only two possibly correct answers: $+2.5 \mu\text{C}/\text{m}^2$ and $+3.2 \mu\text{C}/\text{m}^2$.

So here we have $E = \frac{\sigma_2 - \sigma_1}{2\epsilon_0}$ by choosing the positive direction for the electric field to be from plate 2 to plate 1. Now we need to relate the electric field to the electric-potential difference. Since the electric field is uniform, $\Delta V = E\Delta x$, where Δx is the distance between the two plates. So $E = 2.0 \times 10^3 / 0.01 = 2.0 \times 10^5 \text{ V/m}$. Substitute this into the first equation and solve it for σ_2 to find $\sigma_2 = 2(8.85 \times 10^{-12})(2.0 \times 10^5) - 1.0 \times 10^{-6} = 2.54 \times 10^{-6} \text{ C/m}^2$.

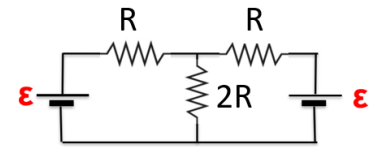
5. Two capacitors, C_1 and C_2 , are separately charged by a 10.0 V battery. Subsequently, the two are arranged as shown, with the positively charged plate of C_1 and the negative charged plate of C_2 being connected to switch S_1 , and the negatively charged plate of C_1 and the positively charged plate of C_2 being connected to switch S_2 . After the two switches are closed, the voltage across capacitor C_1 is found to be 5.0 V. C_2 is a $1.0 \mu\text{F}$ capacitor. What is the capacitance of C_1 , in μF ? Assume that C_1 has a larger capacitance than C_2 .



- (1) ♣ 3.0 (2) 4.0 (3) 2.0 (4) 1.0 (5) 0.0

Before the switches are closed, the charges stored in C_1 and C_2 are respectively $q_1 = C_1\mathcal{E}$ and $q_2 = C_2\mathcal{E}$, where $\mathcal{E} = 10.0 \text{ V}$ is the emf of the battery. After the switches are closed, the total charge stored in the two capacitors will be $q_{\text{total}} = q_1 - q_2$. (Note that $C_1 > C_2$, so $q_1 > q_2$ and q_{total} is $q_1 - q_2$, not $q_2 - q_1$.) As a result, the voltage across either capacitor— C_1 or C_2 —will be $V = q_{\text{total}}/C_{\text{eq}} = (q_1 - q_2)/(C_1 + C_2) = \mathcal{E}(C_1 - C_2)/(C_1 + C_2)$. Plugging in the values $V = 5.0 \text{ V}$ and $\mathcal{E} = 10.0 \text{ V}$ gives $(C_1 - C_2)/(C_1 + C_2) = 0.5$. Solve this for C_1 to find $C_1 = 3.0C_2 = 3.0 \mu\text{F}$.

6. A circuit consists of two ideal batteries of the same emf, and 3 resistors (2 of which are of resistance R and one is of resistance $2R$). What is the current flowing through the resistor of resistance $2R$?

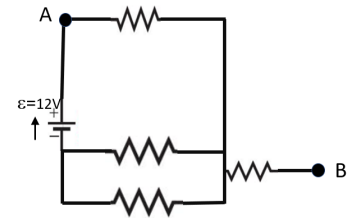


- (1) ♣ $\frac{0.40\epsilon}{R}$ (2) $\frac{0.33\epsilon}{R}$ (3) 0 (4) $\frac{0.66\epsilon}{R}$ (5) $\frac{1.00\epsilon}{R}$

Let us call the current through the central resistor to be i . We call the current exiting the left emf to be i_1 , and the current exiting the right resistor i_2 , we see by symmetry these must be the same. Then Kirchoff's law for junctions finds that $i_1 + i_2 = i$, and so $i_1 = i/2$. The Kirchoff's Law for voltages, using the left loop is $+\epsilon - iR/2 - 2Ri = 0$ And so $i = \frac{2\epsilon}{5R}$

7. Four resistors, each of resistance 6Ω , are attached to a 12 V ideal battery as shown. What is the voltage difference between points A and B, in volts?

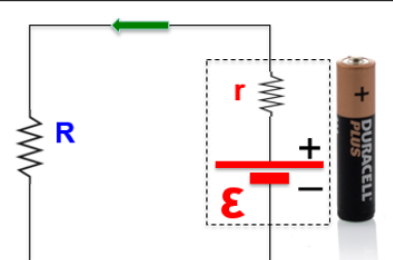
- (1) ♣ 8
(2) 6
(3) 4
(4) 0
(5) 9



Clearly, the resistor next to B is not part of a circuit, so the current through it is zero. Using $V = iR$, that means there is no voltage drop over that resistor. Looking at the main circuit, we have two resistors in parallel which add to give an equivalent resistance of 3Ω . That adds to the resistor at the top to give a total resistance in the circuit of 9Ω . The current through the top resistor is therefore $12/9 = 1.33 \text{ Amps}$. The voltage drop over the top resistor is therefore $6 \times 1.33 = 8\text{V}$, and as there is no drop over the resistor next to B, this is the answer to the question.

8. A non-ideal battery consists of an emf of 1.5 V and an internal resistance of 0.6Ω . It powers an external resistor of resistance 1.0Ω . What is power dissipated in this resistor, in watts?

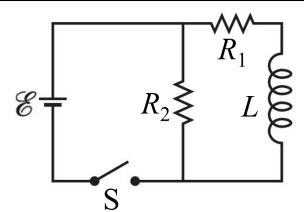
- (1) ♣ 0.88
(2) 2.25
(3) 1.41
(4) 0.94
(5) 1.88



The two resistors are in series giving a total resistance of 1.6Ω and therefore a current of $1.5/1.6 = 0.935 \text{ Amps}$. The power dissipated in the main resistor is $i^2R = 0.935^2 \times 1.0 = 0.88 \text{ W}$.

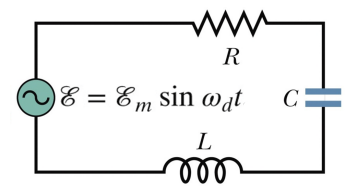
9. In the circuit shown, $\mathcal{E} = 5.0 \text{ V}$, $R_1 = R_2 = 10.0 \Omega$, and $L = 10.0 \text{ mH}$. Switch S is open, and there is no current in any part of the circuit. Right after S is closed, what will be the self-induced emf in the inductor, in volts?

- (1) ♣ 5.0
 (2) 10.0
 (3) 2.5
 (4) infinite
 (5) 0



Immediately after the switch is closed, the inductor acts like broken wire. So the current through it will be zero. This means that the current through R_1 will also be zero, resulting in the voltage drop across the resistor being also zero. For this to happen, the inductor must produce a self-induced emf that exactly cancels the emf of the battery, 5.0 V.

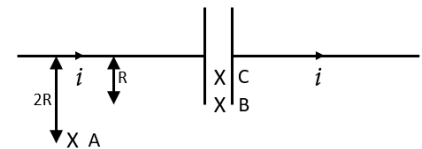
10. A series RLC circuit is driven by a 10 MHz (megahertz) emf source of amplitude $\mathcal{E}_m = 1.0 \text{ V}$. The resistance R is 52Ω , the inductance L is $1.0 \mu\text{H}$, and the capacitance C can be varied. What is the value, in ohms, of the capacitive reactance (X_C) that maximizes the current in the circuit?



- (1) ♣ 63 (2) 52 (3) 0 (4) 27 (5) 86

To maximize the current the circuit must be brought to resonance, at which the capacitive reactance X_C and the inductive reactance X_L cancel each other out in the impedance Z . That is, $X_C = X_L = \omega_d L = 2\pi(1.0 \times 10^7)(1.0 \times 10^{-6}) = 63 \Omega$.

11. The figure shows a capacitor (with circular plates) being charged up and shows three locations, marked by the X symbol, and named A, B and C. The flow of current, i , induces a magnetic field at point A of B_A , where A is a distance of $2R$ away from the wire, and R is the radius of the capacitor plates. The electric field inside the capacitor is uniform. What is the magnetic field at point B, at the edge of the capacitor?



- (1) ♣ $2B_A$ (2) B_A (3) $\frac{B_A}{4}$ (4) 0 (5) $\frac{B_A}{2}$

When a capacitor is charging up, the displacement current between the capacitor plates is the same as the real current charging the capacitor. So, noting that the magnetic field encircling a current is proportion to $1/r$ (where r is the distance from the wire), the distance is halved so the magnetic field is doubled.

12. In the previous question, what is the magnetic field at a point C, a distance of $R/2$ from the center of the capacitor, in terms of the field at point B?

- (1) ♣ $\frac{B_B}{2}$ (2) $2B_B$ (3) B_B (4) $\frac{B_B}{4}$ (5) 0

If you are inside the capacitor, the displacement "current" encircled by the amperian loop is reduced just as current inside an amperian loop is reduced inside a conductor. The displacement current is proportional to the rate of flux change in the capacitor and therefore to the area of the 'amperian loop', and thus to the square of the distance from the center. So we have for the new position: $B 2\pi R/2 = \mu_0 i_{\text{encircled}} = \mu_0 i_{\text{total}} \frac{\pi(R/2)^2}{\pi R^2}$, Thus the answer is half of the B field at the edge of the capacitor.

13. Let u_E and u_B be the densities of energies carried, respectively, by the electric field and the magnetic field of an electromagnetic wave in vacuum. What is the ratio u_E/u_B ? c is the speed of light in vacuum, ϵ_0 the permittivity constant, and μ_0 the permeability constant.

- (1) ♣ 1 (2) The magnetic field carries no energy. (3) 0 (4) c (5) $\sqrt{\epsilon_0 \mu_0}$

The ratio is 1, as was shown in lecture 28.

14. Light polarized along the y direction passes through two polarizing sheets. The polarizing direction of the first sheet is rotated by 45° counterclockwise from the y axis, and the polarizing direction of the second sheet is rotated from the y axis by 15° in the same direction. What is the intensity of the light after the two sheets, if its initial intensity is I_0 ?

- (1) ♣ $\frac{3}{8}I_0$ (2) 0 (3) I_0 (4) $0.467I_0$ (5) $\frac{1}{2}I_0$

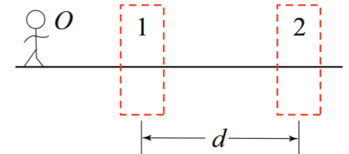
A polarizing sheet reduces the intensity of polarized light, with initial intensity I_0 , to $I_0 \cos^2 \theta$, where θ is the angle between the polarization direction of the light and the polarizing direction of the sheet. The first polarizing sheet reduces the intensity to $I_1 = I_0 \cos^2 45^\circ$, and the second sheet reduces the intensity from I_1 to $I_2 = I_1 \cos^2(45^\circ - 15^\circ)$. (45° is the direction of the polarization of the light after the first sheet, and 15° is the polarizing direction of the second sheet. The angle between the two directions is θ .) Combining the two equations, we get $I_2 = I_0 \cos^2 45^\circ \cos^2 30^\circ = I_0 (\frac{1}{\sqrt{2}})^2 (\frac{\sqrt{3}}{2})^2 = \frac{3}{8}I_0$.

15. A spherical mirror produces an image of an object, which is located 10 cm in front of the mirror. (The object is on the central axis of the mirror.) The height of the image is 3 times that of the object. What is the focal length (ignore the sign) of the mirror? Is the mirror convex or concave?

- (1) ♣ 15 cm or 7.5 cm, concave
 (2) 15 cm or 7.5 cm, convex
 (3) 15 cm, concave (cannot be 7.5 cm)
 (4) 7.5 cm, concave (cannot be 15 cm)
 (5) 15 cm, convex (cannot be 7.5 cm)

A convex mirror can produce only a virtual image whose height is smaller, not larger, than that of the object. So the mirror must be concave. The problem does not specify whether the image is real or virtual. If real, $m = -i/p = -3$, where m is the magnification, i the image distance, and $p = 10$ cm the object distance. This gives $i = 30$ cm. Plug the p and i values into $\frac{1}{p} + \frac{1}{i} = \frac{1}{f}$ to find the focal length $f = 7.5$ cm. If the image is virtual, on the other hand, then $m = -i/p = +3$, hence $i = -30$ cm. This i value, together with $p = 10$ cm, gives $f = 15$ cm.

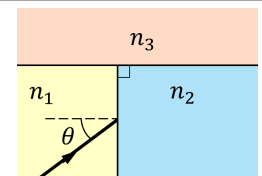
16. In the figure, which is not to scale, boxes 1 and 2 represent lenses with a common central axis, and stick figure O represents an object standing on that axis. Lens 1 is converging, with a 10 cm focal length, and lens 2 is diverging, with a 15 cm focal length. The object is located 20 cm to the left of lens 1, and the distance d between the two lenses is 10 cm. Where is the final image produced by the two-lens system? Is the image real or virtual?



- (1) ♣ 30 cm to the right of lens 2, real
 (2) 20 cm to the left of lens 2, virtual
 (3) 20 cm to the right of lens 2, virtual
 (4) 10 cm to the right of lens 2, real
 (5) 10 cm to the left of lens 2, real

For lens 1, $\frac{1}{20} + \frac{1}{i_1} = \frac{1}{10}$, where i_1 is the distance, from lens 1, of the image that lens 1 would produce if lens 2 were absent. The equation gives $i_1 = 20$ cm. This image acts as an object for lens 2, located $20 \text{ cm} - d = 10$ cm to the right of lens 2. For lens 2, therefore, $\frac{1}{-10} + \frac{1}{i_2} = \frac{1}{-15}$, where i_2 is the distance of the final image from lens 2. (The object distance needs the negative sign because it is on the "wrong" side of the lens.) This gives $i_2 = 30$ cm. Since the value is positive, the image is real and located to the right of lens 2.

17. A ray of light is incident on a vertical interface between materials 1 and 2, as shown. What is the maximum value of the incident angle θ that causes the ray refracted into medium 2 to undergo total internal reflection at the horizontal interface between materials 2 and 3? Materials 1, 2, and 3 have indices of refraction $n_1 = 2.0$, $n_2 = 1.6$, and $n_3 = 1.4$, respectively.



- (1) ♣ 23° (2) 36° (3) 40° (4) 51° (5) 62°

As θ increases, the incident angle, θ_i , of the refracted ray at the interface between materials 2 and 3 will decrease, eventually reaching the critical angle θ_c below which the ray will not be totally reflected at the interface. So we just need to find θ that makes $\theta_i = \theta_c$. θ_c is given by $n_2 \sin \theta_c = n_3 \cdots (1)$. When $\theta_i = \theta_c$, the refraction angle at the interface between materials 1 and 2 is $\pi/2 - \theta_c$. So $n_1 \sin \theta = n_2 \sin(\pi/2 - \theta_c)$, that is $n_1 \sin \theta = n_2 \cos \theta_c \cdots (2)$. To find θ , you may square Eqs. 1 and 2 and add them to eliminate θ_c , obtaining $n_1^2 \sin^2 \theta + n_2^2 = n_2^2$. This gives $\sin \theta = \frac{\sqrt{n_2^2 - n_3^2}}{n_1} = \frac{\sqrt{0.6}}{2.0} = 0.3873$, which in turn gives $\theta = 23^\circ$.

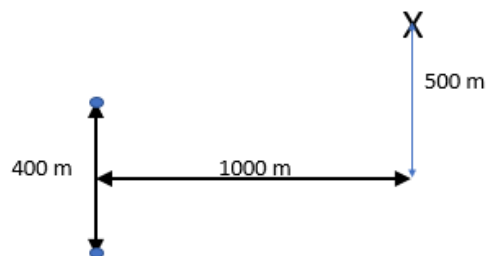
Alternatively, you may insert the values of n_2 and n_3 into Eq. 1, solve it for θ_c , substitute that into Eq. 2, and solve it for θ .

18. An incident light wave of wavelength $\lambda = 500 \text{ nm}$ is incident on a single slit of width 1.0 mm . It is viewed on a screen 2 meters away from the slit. What is the distance between the center of the pattern on the screen and the first minimum (dark line)?

- (1) ♣ 1.0 mm (2) 0.5 mm (3) 2.0 mm (4) 4.0 mm (5) 3.0 mm

For a single slit the angle of the first minimum from the center is given by $a \sin\theta = m\lambda$ where we are given λ and the slit width a . This needs to be multiplied by the distance from the screen (2 meters) to find the distance on the screen from the center of the pattern to the first minimum (note that the approximation $\sin\theta = \tan\theta = \theta$ is a very good one).

19. A radio station sends identical signals to two antenna towers that are 400 meters apart as shown. A car, traveling along a straight road parallel to the line joining the towers and 1000 m away from it, finds a first minimum in reception when it is at a point X, 500 m from the point immediately opposite the antennas. What is the radio station's broadcasting wavelength, in meters?



- (1) ♣ 350 m (2) 200 m (3) 270 m (4) 800 m (5) 400 m

We cannot use the equation for two slits to answer this, as the angles are really big and so the approximations that make that equation are not valid. Instead we use the Pythagorean formula to find that the distance from one antenna to the car is $\sqrt{1000^2 + 300^2} = 1044$ meters, and from the other is $\sqrt{1000^2 + 700^2} = 1220$ meters. The difference is 176 meters, which, for the first minimum, must be half a wavelength.

20. Two light rays are in phase and have a wavelength of 700 nm . One continues in air and the second travels through a very thin plastic layer of index of refraction $n = 1.6$ and thickness $1.0 \mu\text{m}$. After emerging, what is their phase difference in units of their wavelength?

- (1) 0.43 (2) 0.30 (3) 0.60 (4) 0.21 (5) ♣ 0.85

The number of wavelengths for L length in air is L/λ , whereas in the plastic it is nL/λ . So the difference is $\frac{(1.6-1.0)1.0 \times 10^{-6}}{7 \times 10^{-7}} = 0.85$ wavelengths.

Scratch paper

Scratch paper

Scratch paper

Formulas – page 1

Constants: $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$ $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2$.

$e = 1.6 \times 10^{-19} \text{ C}$ $k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$.

$c = 3.0 \times 10^8 \text{ m/s}$

kilo (k) = 10^3 ; centi (c) = 10^{-2} ; milli (m) = 10^{-3} ; micro (μ) = 10^{-6} ; nano (n) = 10^{-9} ; pico (p) = 10^{-12}

Coulomb's Law: $F = \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{r^2}$

Electric field: $\vec{F} = q\vec{E}$ $E = \frac{qz}{4\pi\epsilon_0(z^2 + R^2)^{3/2}}$ (charged ring)

$E = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2}$ (point charge) $E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}}\right)$ (charged disk)

$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2}$ (continuous) $E = \frac{\sigma}{2\epsilon_0}$ (infinite sheet)

Dipoles: $p = qd$ $\tau = pE \sin \theta$ $U = -pE \cos \theta$

Gauss' Law: $\Phi = \int \vec{E} \cdot d\vec{A} = (E \cos \theta)A$ $E = \frac{\sigma}{\epsilon_0}$ (conducting surface)

$\epsilon_0 \Phi = \epsilon_0 \oint \vec{E} \cdot d\vec{A} = q_{\text{enc}}$ $E = \frac{\lambda}{2\pi\epsilon_0 r}$ (line of charge)

Electric potential: $U = qV$ $V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$ (point charge)

$V_f - V_i = -\int_i^f \vec{E} \cdot d\vec{s}$ $dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r}$ (continuous)

$E_x = -\frac{\partial V}{\partial x}; E_y = -\frac{\partial V}{\partial y}; E_z = -\frac{\partial V}{\partial z}$ $U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$ (two-particle system)

Capacitors: $q = CV$ $C = \frac{\epsilon_0 A}{d}$ (parallel plate)

$C_{\text{eq}} = \sum_{j=1}^n C_j$ (n capacitors in parallel) $C = 2\pi\epsilon_0 \frac{L}{\ln(b/a)}$ (cylindrical)

$\frac{1}{C_{\text{eq}}} = \sum_{j=1}^n \frac{1}{C_j}$ (n capacitors in series) $C = 4\pi\epsilon_0 \frac{ab}{b-a}$ (spherical)

$U_E = \frac{q^2}{2C} = \frac{1}{2} CV^2$ $C = 4\pi\epsilon_0 R$ (isolated sphere)

$u_E = \frac{1}{2} \epsilon_0 E^2$ $C = \kappa \frac{\epsilon_0 A}{d}$ (parallel plate with dielectric)

Formulas – page 2

Current:	$i = \frac{dq}{dt},$	$i = \int \vec{J} \cdot d\vec{A},$	$\vec{J} = (ne)\vec{v}_d,$	
Resistors:	$R = \frac{V}{i},$	$v_R = iR.$	$R = \rho \frac{L}{A},$	$\rho = \frac{1}{\sigma} = \frac{E}{J},$
	$\frac{1}{R_{eq}} = \sum_{j=1}^n \frac{1}{R_j}$	$(n \text{ resistances in parallel}).$		$R_{eq} = \sum_{j=1}^n R_j$ $(n \text{ resistances in series}).$
Inductors:	$L = \frac{N\Phi_B}{i}$	$\mathcal{E}_L = -L \frac{di}{dt}.$	$\frac{L}{l} = \mu_0 n^2 A$	$(\text{solenoid}).$
	Inductors add in series and parallel with the same rules as resistors.			$\mathcal{E}_2 = -M \frac{di_1}{dt}$
	$U_B = \frac{Li^2}{2},$	$u_B = \frac{B^2}{2\mu_0}$	$\mathcal{E}_1 = -M \frac{di_2}{dt},$	
RC circuit:	$q = C\mathcal{E}(1 - e^{-t/\tau_C}).$	(charging)	$q = q_0 e^{-t/\tau_C}.$	(discharging)
			$\tau_C = RC.$	
RL circuit:	$i = \frac{\mathcal{E}}{R}(1 - e^{-t/\tau_L})$	$(\text{rise of current}).$	$i = i_0 e^{-t/\tau_L}$	$(\text{decay of current}).$
			$\tau_L = \frac{L}{R}$	
LC circuit:	$q = Q \cos(\omega t + \phi)$	$(\text{charge}),$	$i = \frac{dq}{dt} = -\omega Q \sin(\omega t + \phi)$	$(\text{current}).$
			$\omega = \frac{1}{\sqrt{LC}}.$	
RLC circuit: (not driven)	$q = Qe^{-Rt/2L} \cos(\omega' t + \phi),$		$\omega' = \sqrt{\omega^2 - (R/2L)^2}.$	
Driven circuits:	$\mathcal{E} = \mathcal{E}_m \sin \omega_d t.$			
One circuit element:	R		$i_R = \frac{v_R}{R} = \frac{V_R}{R} \sin \omega_d t.$	
	$X_C = \frac{1}{\omega_d C}$	$(\text{capacitive reactance}).$	$i_C = \left(\frac{V_C}{X_C}\right) \sin(\omega_d t + 90^\circ).$	
	$X_L = \omega_d L$	$(\text{inductive reactance}).$	$i_L = \left(\frac{V_L}{X_L}\right) \sin(\omega_d t - 90^\circ).$	
Series RLC:	$Z = \sqrt{R^2 + (X_L - X_C)^2}$	(impedance)	$i = I \sin(\omega_d t - \phi),$	
	$\tan \phi = \frac{X_L - X_C}{R}$	$(\text{phase constant}).$	$I = \frac{\mathcal{E}_m}{\sqrt{R^2 + (X_L - X_C)^2}}.$	
Power:	$I_{rms} = \frac{I}{\sqrt{2}}$	$\mathcal{E}_{rms} = \frac{\mathcal{E}_m}{\sqrt{2}}$	$P_{avg} = \mathcal{E}_{rms} I_{rms} \cos \phi$	
Transformers:	$V_s = V_p \frac{N_s}{N_p}$	$I_s = I_p \frac{N_p}{N_s}$	$R_{eq} = \left(\frac{N_p}{N_s}\right)^2 R,$	

Formulas – page 3

Forces: $\vec{F} = q\vec{E}$, $\vec{F}_B = q\vec{v} \times \vec{B}$; $F_B = |q|vB \sin \phi$, $\vec{F}_B = i\vec{L} \times \vec{B}$. $F_{ba} = \frac{\mu_0 L_a i_b}{2\pi d}$

Circular motion: $|q|vB = \frac{mv^2}{r}$, $r = \frac{mv}{|q|B}$. $f = \frac{\omega}{2\pi} = \frac{1}{T} = \frac{|q|B}{2\pi m}$. Hall effect: $n = \frac{Bi}{Vle}$,

Magnetic moment: $\mu = NiA$ $\vec{\tau} = \vec{\mu} \times \vec{B}$. $\tau = NiAB \sin \theta$, $U(\theta) = -\vec{\mu} \cdot \vec{B}$.

Magnetic field: $d\vec{B} = \frac{\mu_0}{4\pi} \frac{i d\vec{s} \times \hat{r}}{r^2}$ (Biot–Savart law) $B = \frac{\mu_0 i}{2\pi R}$ (long straight wire)

$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{\text{enc}}$ (Ampere's law) $B = \frac{\mu_0 i \phi}{4\pi R}$ (at center of circular arc)

$B = \frac{\mu_0 i (2\pi)}{4\pi R} = \frac{\mu_0 i}{2R}$ (at center of full circle)

$B = \mu_0 i n$ (ideal solenoid), $B = \frac{\mu_0 i N}{2\pi} \frac{1}{r}$ (toroid)

Faraday's law: $\mathcal{E} = \oint \vec{E} \cdot d\vec{s}$. $\mathcal{E} = -\frac{d\Phi_B}{dt}$ $\Phi_B = \int \vec{B} \cdot d\vec{A}$

Maxwell's eqs.: Gauss' law for electricity $\oint \vec{E} \cdot d\vec{A} = q_{\text{enc}}/\epsilon_0$

Gauss' law for magnetism $\oint \vec{B} \cdot d\vec{A} = 0$

Faraday's law $\oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$

Ampere–Maxwell law $\oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} + \mu_0 i_{\text{enc}}$ $i_d = \epsilon_0 \frac{d\Phi_E}{dt}$.

Waves: $A = A_m \sin(kx - \omega t)$ $k = \frac{2\pi}{\lambda}$ $\omega = \frac{2\pi}{T}$ $c = f\lambda = \frac{\omega}{k}$

Electromagnetic waves:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$\frac{E}{B} = c$$

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$$

$$I = \frac{1}{2\mu_0 c} E_m^2$$

$$p_r = \frac{I}{c} \quad (\text{total absorption})$$

$$p_r = \frac{2I}{c} \quad (\text{total reflection back along path}).$$

Formulas – page 4

Polarizers: $I = \frac{1}{2} I_0.$ $I = I_0 \cos^2 \theta.$

Refraction: $n_2 \sin \theta_2 = n_1 \sin \theta_1,$ $\theta_c = \sin^{-1} \frac{n_2}{n_1}$ (critical angle).

Polarization by reflection: $\theta_B = \tan^{-1} \frac{n_2}{n_1}$ (Brewster angle).

Mirrors: $i = -p$ (plane mirror). $\frac{1}{p} + \frac{1}{i} = \frac{1}{f} = \frac{2}{r}.$ $m = -\frac{i}{p}.$

Lenses: $\frac{1}{p} + \frac{1}{i} = \frac{1}{f}$ (thin lens), $m = -\frac{i}{p}$

Two slits: $d \sin \theta = m\lambda,$ for $m = 0, 1, 2, \dots$ (maxima—bright fringes), $d \sin \theta = (m + \frac{1}{2})\lambda,$ for $m = 0, 1, 2, \dots$ (minima—dark fringes),

Single slit: $a \sin \theta = m\lambda,$ for $m = 1, 2, 3, \dots$ (minima).

Telescope angular resolution: $\Delta\theta = 1.22 \frac{\lambda}{D}$

Diffraction grating: $d \sin \theta_m^{\max} = m\lambda$

Details of Master Exam Questions

Total of 19 QQ, 1 CQ, 0 RQ questions (0 distinct RQ questions)

* Total of 20 questions defined (20 for each student exam)

* Total weight is 20.00 for each student exam

* Total of 15 figures defined

Shuffle								Short text
Q	Type	aType	Wght	#fig	#ans	ans?	Correct	
1	QQ	Single	1.00	1	5	Yes	1	A small electric dipole is placed in a uniform electric
2	QQ	Single	1.00	0	5	Yes	1	Two charges, $+Q$ and $-2Q$, are placed on the x -axis at
3	QQ	Single	1.00	0	5	Yes	1	A long cylindrical shell of outer radius 4.0 cm and in
4	QQ	Single	1.00	0	5	Yes	1	Two parallel, infinite, non-conducting planes, 1 and 2,
5	QQ	Single	1.00	1	5	Yes	1	Two capacitors, C_1 and C_2 , are separately charged
6	QQ	Single	1.00	1	5	Yes	1	A circuit consists of two ideal batteries of the same e
7	QQ	Single	1.00	1	5	Yes	1	Four resistors, each of resistance $6\ \Omega$, are atta
8	QQ	Single	1.00	1	5	Yes	1	A non-ideal battery consists of an emf of 1.5 V and an
9	QQ	Single	1.00	1	5	Yes	1	In the circuit shown, $\mathcal{E} = 5.0\text{ V}$, R_1
10	QQ	Single	1.00	1	5	Yes	1	A series RLC circuit is driven by a 10 MHz (megahertz)
11	QQ	Single	1.00	1	5	Yes	1	The figure shows a capacitor (with circular plates) bei
12	CQ	Single	1.00	0	5	Yes	1	In the previous question, what is the magnetic field at
13	QQ	Single	1.00	0	5	Yes	1	Let u_E and u_B be the densities of energies carrie
14	QQ	Single	1.00	0	5	Yes	1	Light polarized along the y direction passes through
15	QQ	Single	1.00	0	5	Yes	1	A spherical mirror produces an image of an object, whic
16	QQ	Single	1.00	1	5	Yes	1	In the figure, which is not to scale, boxes 1 and 2 rep
17	QQ	Single	1.00	1	5	Yes	1	A ray of light is incident on a vertical interface betw
18	QQ	Single	1.00	0	5	Yes	1	An incident light wave of wavelength $\lambda = 500\text{ nm}$
19	QQ	Single	1.00	1	5	Yes	1	A radio station sends identical signals to two antenna
20	QQ	Single	1.00	0	5	Yes	5	Two light rays are in phase and have a wavelength of